

ENGINEERING MECHANICS

UNIT IV: CENTROID AND MOMENT OF INERTIA

Harcourt Butler Technical University (HBTU), Kanpur — B.Tech 1st Year

Centroids of Curves, Areas, Volumes & Composite Sections, Area MOI, Parallel/Perpendicular Theorems, Principal Axes, and Mass MOI of Symmetric Bodies

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1 Part 1: Centroid and Center of Gravity

1.1 1.1 Definitions and Fundamental Distinctions

Core Terminology

- **Center of Gravity (C.G.):** The single point through which the resultant gravitational force (total weight W) of a body acts, regardless of the body's orientation in space.

$$\bar{x} = \frac{\int x dW}{W}, \quad \bar{y} = \frac{\int y dW}{W}, \quad \bar{z} = \frac{\int z dW}{W} \quad (1)$$

- **Center of Mass (C.M.):** The point where the total mass M of a system is concentrated.

In a uniform gravitational field ($g = \text{constant}$), the Center of Gravity coincides identically with the Center of Mass.

- **Centroid (C):** The purely **geometrical center** of an idealized geometric figure (line curve, plane area, or solid volume). It depends strictly on geometry and boundary shape, completely independent of mass density and gravity. If a body has uniform, homogeneous density, its Centroid, Center of Mass, and Center of Gravity are all coincident.

1.2 Centroids of Standard Geometries

1.2.1 A. Centroid of a Plane Curve / Line (L)

For a one-dimensional curve of total arc length $L = \int dL$:

$$\bar{x} = \frac{\int x dL}{L}, \quad \bar{y} = \frac{\int y dL}{L} \quad (2)$$

- **Semicircular Arc of Radius R :** Centered at origin, symmetric about y -axis:

$$\bar{x} = 0, \quad \bar{y} = \frac{2R}{\pi} \approx 0.6366R \quad (3)$$

- **Quarter-Circular Arc of Radius R :** $\bar{x} = \frac{2R}{\pi}$, $\bar{y} = \frac{2R}{\pi}$.

1.2.2 B. Centroid of a Plane Area (A)

For a two-dimensional lamina of total surface area $A = \int dA$:

$$\bar{x} = \frac{\int x dA}{A} = \frac{Q_y}{A}, \quad \bar{y} = \frac{\int y dA}{A} = \frac{Q_x}{A} \quad (4)$$

where $Q_y = \int x dA$ is the **First Moment of Area** with respect to the y -axis, and $Q_x = \int y dA$ is with respect to the x -axis. If an axis is an axis of geometric symmetry, the first moment of area about that axis is zero, and the centroid lies on that axis!

Geometric Shape	Area (A)	Centroid Coordinates ($\bar{x}, \bar{y}$)	Reference Datum
Rectangle ($b \times h$)	$A = b \cdot h$	$\bar{x} = b/2, \quad \bar{y} = h/2$	Bottom-left corner
Right Triangle (base b , height h)	$A = \frac{1}{2}bh$	$\bar{x} = b/3, \quad \bar{y} = h/3$	At right-angle vertex
General Triangle (base b , height h)	$A = \frac{1}{2}bh$	$\bar{y} = h/3$	Measured from base
Circle (radius R)	$A = \pi R^2$	At center ($\bar{x} = 0, \bar{y} = 0$)	Center of circle
Semicircular Area (radius R)	$A = \frac{\pi R^2}{2}$	$\bar{x} = 0, \quad \bar{y} = \frac{4R}{3\pi} \approx 0.4244R$	Flat base center (on axis of sym.)
Quarter-Circle Area (radius R)	$A = \frac{\pi R^2}{4}$	$\bar{x} = \frac{4R}{3\pi}, \quad \bar{y} = \frac{4R}{3\pi}$	Corner vertex center
Parabolic Spandrel ($y = kx^2$)	$A = \frac{1}{3}bh$	$\bar{x} = \frac{3}{4}b, \quad \bar{y} = \frac{3}{10}h$	Vertex origin $(0, 0)$

Table 1: Centroid Locations for Standard 2D Plane Areas

1.2.3 C. Centroid of Solid Volumes (V)

For a three-dimensional solid volume $V = \int dV$:

$$\bar{x} = \frac{\int x dV}{V}, \quad \bar{y} = \frac{\int y dV}{V}, \quad \bar{z} = \frac{\int z dV}{V} \quad (5)$$

- **Right Circular Cone / Pyramid (height h):** $\bar{z} = \frac{h}{4}$ (measured from circular base along symmetry axis).
- **Solid Hemisphere (radius R):** $\bar{z} = \frac{3R}{8}$ (measured from flat circular face along symmetry axis).
- **Hemispherical Thin Shell (radius R):** $\bar{z} = \frac{R}{2}$ (measured from base circle).

1.3 Centroid of Composite Bodies

Structural members (I-sections, T-sections, channels, angle irons, bracket plates with cutouts) are composed of connected standard sub-shapes (A_i or V_i).

Composite Area Centroid Formulas

$$\bar{X} = \frac{\sum A_i \bar{x}_i}{\sum A_i} = \frac{A_1 \bar{x}_1 + A_2 \bar{x}_2 + \cdots + A_n \bar{x}_n}{A_1 + A_2 + \cdots + A_n} \quad (6)$$

$$\bar{Y} = \frac{\sum A_i \bar{y}_i}{\sum A_i} = \frac{A_1 \bar{y}_1 + A_2 \bar{y}_2 + \cdots + A_n \bar{y}_n}{A_1 + A_2 + \cdots + A_n} \quad (7)$$

Holes and Cutouts Rule: For any negative cutout or hollow hole, treat its area as **negative** (i.e., subtract its component: $A_{\text{hole}} < 0$).

2 Part 2: Moment of Inertia of Plane Areas (Area MOI)

2.1 Definition and Physical Significance

The **Second Moment of Area** (commonly termed the Area Moment of Inertia) is a purely geometrical property of a structural cross-section that measures its resistance to bending deformation and deflection about a specified axis.

- About x -axis:

$$I_{xx} = \int_A y^2 dA \quad (\text{units: mm}^4, \text{m}^4) \quad (8)$$

- About y -axis:

$$I_{yy} = \int_A x^2 dA \quad (9)$$

- Polar Moment of Inertia (J or I_z):** Second moment of area about an axis perpendicular to the plane passing through origin O :

$$J_O = \int_A r^2 dA = \int_A (x^2 + y^2) dA = I_{xx} + I_{yy} \quad (10)$$

- Radius of Gyration (k):** The imaginary distance from the reference axis at which the entire area could be concentrated into a strip without changing its second moment:

$$k_{xx} = \sqrt{\frac{I_{xx}}{A}}, \quad k_{yy} = \sqrt{\frac{I_{yy}}{A}}, \quad k_z = \sqrt{\frac{J}{A}} \quad (11)$$

2.2 Fundamental Theorems of Moment of Inertia

2.2.1 A. Parallel Axes Theorem (Steiner's Theorem)

Parallel Axes Theorem Statement

The moment of inertia of any plane area about an arbitrary axis AB is equal to the moment of inertia of the area about a **mutually parallel centroidal axis** GG passing through its centroid plus the product of the area and the square of the perpendicular distance d between the two axes.

$$I_{AB} = I_G + Ad^2 \quad (12)$$

Rigid Condition: One of the two parallel axes **must** pass through the area's own centroid G !

2.2.2 B. Perpendicular Axes Theorem

Perpendicular Axes Theorem Statement

For any plane lamina lying in the xy -plane, the moment of inertia about an axis z perpendicular to the plane (polar axis) is equal to the sum of its moments of inertia about any two mutually perpendicular coplanar axes x and y intersecting at the same origin:

$$\mathbf{I}_{zz} = \mathbf{J}_O = \mathbf{I}_{xx} + \mathbf{I}_{yy} \quad (13)$$

Limitation: Valid strictly for **two-dimensional plane laminas**; cannot be applied to 3D volumetric bodies!

2.3 2.3 Standard Area Moment of Inertia Table

Cross-Section	Centroidal MOI ($I_{G,xx}$)	Centroidal MOI ($I_{G,yy}$)	Base / Edge MOI
Rectangle ($b \times h$)	$I_{Gx} = \frac{bh^3}{12}$	$I_{Gy} = \frac{hb^3}{12}$	$I_{\text{base}} = \frac{bh^3}{3}$
Triangle (base b , height h)	$I_{Gx} = \frac{bh^3}{36}$	—	$I_{\text{base}} = \frac{bh^3}{12}, \quad I_{\text{apex}} = \frac{bh^3}{4}$
Circle (radius R , dia d)	$I_{Gx} = \frac{\pi R^4}{4} = \frac{\pi d^4}{64}$	$I_{Gy} = \frac{\pi R^4}{4} = \frac{\pi d^4}{64}$	$J_O = \frac{\pi R^4}{2} = \frac{\pi d^4}{32}$
Semicircle (radius R)	$I_{Gx} = \left(\frac{\pi}{8} - \frac{8}{9\pi}\right) R^4 \approx 0.1098R^4$	$I_{Gy} = \frac{\pi R^4}{8}$	$I_{\text{diametral base}} = \frac{\pi R^4}{8}$
Quarter-Circle (radius R)	$I_{Gx} \approx 0.05488R^4$	$I_{Gy} \approx 0.05488R^4$	$I_{\text{corner edge}} = \frac{\pi R^4}{16}$

Table 2: Standard Second Moments of Area

2.4 2.4 Product of Inertia and Principal Moments of Inertia

- **Product of Inertia (I_{xy}):**

$$I_{xy} = \int_A xy \, dA \quad (14)$$

Unlike I_{xx} and I_{yy} (which are always strictly positive), I_{xy} can be **positive, negative, or zero**.

- **Symmetry Theorem:** If a cross-section has at least one axis of geometric symmetry, its product of inertia with respect to that symmetry axis and any axis perpendicular to it is identically **zero**:

$$\text{Axis of symmetry present} \implies I_{xy} = 0 \quad (15)$$

- **Parallel Axes Theorem for Product of Inertia:**

$$I_{xy} = I_{\bar{x}\bar{y}} + A \bar{x}_G \bar{y}_G \quad (16)$$

Transformation of Axes & Principal Moments of Inertia

When coordinate axes are rotated by an angle θ (counter-clockwise):

$$I_u = \frac{I_{xx} + I_{yy}}{2} + \frac{I_{xx} - I_{yy}}{2} \cos 2\theta - I_{xy} \sin 2\theta \quad (17)$$

$$I_{uv} = \frac{I_{xx} - I_{yy}}{2} \sin 2\theta + I_{xy} \cos 2\theta \quad (18)$$

Principal Axes: The particular orientation of orthogonal axes (u, v) for which the product of inertia vanishes identically ($I_{uv} = 0$). The angle of inclination θ_p is given by:

$$\tan 2\theta_p = -\frac{2I_{xy}}{I_{xx} - I_{yy}} \quad (19)$$

Principal Moments of Inertia ($I_{\max}, I_{\min}$):

$$I_{1,2} = \frac{I_{xx} + I_{yy}}{2} \pm \sqrt{\left(\frac{I_{xx} - I_{yy}}{2}\right)^2 + I_{xy}^2} \quad (20)$$

Note on Invariant Property: $I_{xx} + I_{yy} = I_1 + I_2 = J_O$ remains constant under all rotations.

3 Part 3: Mass Moment of Inertia

3.1 3.1 Definition and Physical Concept

Mass Moment of Inertia (I_m or I) is the measure of a rigid body's rotational inertia (reluctance to angular acceleration α) about a given axis:

$$I = \int r^2 dm = \sum m_i r_i^2 \quad (\text{units: kg} \cdot \text{m}^2) \quad (21)$$

Rotational counterpart to Newton's second law: $T = I\alpha$.

- **Radius of Gyration (k_m):** $k = \sqrt{\frac{I}{M}} \implies I = Mk^2$.
- **Parallel Axis Theorem for Mass MOI:**

$$I = I_G + Md^2 \quad (22)$$

where I_G is about the parallel axis passing through the Center of Mass G , and d is the distance between axes.

3.2 3.2 Mass Moment of Inertia of Standard Bodies About Axis of Symmetry

Catalog of Standard Symmetric Bodies (Mass M)

1. **Thin Circular Ring / Thin Cylindrical Shell (Radius R):** All elemental mass particles lie at identical distance R from the central longitudinal symmetry axis:

$$I_{zz} = MR^2, \quad I_{xx} = I_{yy} = \frac{1}{2}MR^2 \quad (\text{about diametral axis}) \quad (23)$$

2. **Uniform Circular Disc / Solid Cylinder (Radius R , length L):** Integrating concentric cylindrical shells of radius r ($dm = \rho 2\pi r L dr$):

$$I_{zz} = \int_0^R r^2 (\rho 2\pi r L dr) = 2\pi \rho L \left[\frac{r^4}{4} \right]_0^R = \frac{1}{2}(\rho \pi R^2 L) R^2 \implies I_{zz} = \frac{1}{2}MR^2 \quad (24)$$

About transverse centroidal axis: $I_{xx} = I_{yy} = \frac{1}{4}MR^2 + \frac{1}{12}ML^2$.

3. **Solid Sphere (Radius R):** Taking thin disc elements perpendicular to diameter:

$$I_{\text{diameter}} = \frac{2}{5}MR^2 \quad (25)$$

Thin Spherical Shell: $I_{\text{diameter}} = \frac{2}{3}MR^2$.

4. **Solid Right Circular Cone (Base radius R , height h):** About central longitudinal axis of revolution (apex-to-base centerline):

$$I_{zz} = \frac{3}{10}MR^2 \quad (26)$$

About transverse centroidal axis (at $h/4$ from base): $I_{xx} = I_{yy} = \frac{3}{20}MR^2 + \frac{3}{80}Mh^2$.

4 Part 4: Representative Analytical Solved Frameworks

4.1 Problem 1: Centroid and MOI of an Asymmetric T-Section

Problem Statement: A T-section consists of a top flange of dimensions $120 \text{ mm} \times 20 \text{ mm}$ and a vertical web of dimensions $20 \text{ mm} \times 140 \text{ mm}$ (total height 160 mm). Determine:

1. Location of centroid $(\bar{X}, \bar{Y})$ from bottom edge.
2. Centroidal moments of inertia $I_{\bar{x}\bar{x}}$ and $I_{\bar{y}\bar{y}}$.

Step-by-step Solution:

1. **Decomposition and Datum:** Place the origin $(0, 0)$ at the bottom-left corner of the web. The section is symmetric about the central vertical line $\Rightarrow \bar{\mathbf{X}} = \mathbf{0}$ (on centerline).

- Web (Part 1): $b_1 = 20 \text{ mm}$, $h_1 = 140 \text{ mm} \Rightarrow A_1 = 20 \times 140 = 2800 \text{ mm}^2$. Centroid $\bar{y}_1 = 140/2 = 70 \text{ mm}$.
- Flange (Part 2): $b_2 = 120 \text{ mm}$, $h_2 = 20 \text{ mm} \Rightarrow A_2 = 120 \times 20 = 2400 \text{ mm}^2$. Centroid $\bar{y}_2 = 140 + (20/2) = 150 \text{ mm}$.

2. **Centroid Calculation ($\bar{Y}$):**

$$\begin{aligned}\bar{Y} &= \frac{A_1 \bar{y}_1 + A_2 \bar{y}_2}{A_1 + A_2} = \frac{(2800 \times 70) + (2400 \times 150)}{2800 + 2400} \\ &= \frac{196\,000 + 360\,000}{5200} = \frac{556\,000}{5200} \approx \mathbf{106.92 \text{ mm}} \quad (\text{from bottom edge})\end{aligned}$$

3. **Centroidal Moment of Inertia $I_{\bar{x}\bar{x}}$ (Parallel Axis Theorem):**

- Distance $d_1 = |\bar{Y} - \bar{y}_1| = |106.92 - 70| = 36.92 \text{ mm}$.
- Distance $d_2 = |\bar{Y} - \bar{y}_2| = |106.92 - 150| = 43.08 \text{ mm}$.
- For Web:

$$I_{x1} = \frac{b_1 h_1^3}{12} + A_1 d_1^2 = \frac{20 \times 140^3}{12} + 2800 \times (36.92)^2 = 4\,573\,333 + 3\,816\,644 = 8\,389\,977 \text{ mm}^4$$

- For Flange:

$$I_{x2} = \frac{b_2 h_2^3}{12} + A_2 d_2^2 = \frac{120 \times 20^3}{12} + 2400 \times (43.08)^2 = 80\,000 + 4\,454\,134 = 4\,534\,134 \text{ mm}^4$$

- Total $I_{\bar{x}\bar{x}}$:

$$\mathbf{I_{\bar{x}\bar{x}} = I_{x1} + I_{x2} = 8\,389\,977 + 4\,534\,134 = 12.924 \times 10^6 \text{ mm}^4}$$

4. **Centroidal Moment of Inertia $I_{\bar{y}\bar{y}}$:** Since the vertical axis passes through the centroid of both sub-areas ($d_x = 0$):

$$\mathbf{I_{\bar{y}\bar{y}} = \frac{h_1 b_1^3}{12} + \frac{h_2 b_2^3}{12} = \frac{140 \times 20^3}{12} + \frac{20 \times 120^3}{12} = 93\,333 + 2\,880\,000 = 2.973 \times 10^6 \text{ mm}^4}$$

4.2 Problem 2: Principal Moments of Inertia of an Unequal Angle Section

Problem Statement: For an unequal angle section $100 \text{ mm} \times 80 \text{ mm}$, the calculated centroidal moments and product of inertia are $I_{xx} = 2.45 \times 10^6 \text{ mm}^4$, $I_{yy} = 1.35 \times 10^6 \text{ mm}^4$, and $I_{xy} = -1.05 \times 10^6 \text{ mm}^4$. Determine the principal moments of inertia and the orientation of the principal axes.

Step-by-step Solution:

1. Orientation of Principal Axes:

$$\tan 2\theta_p = -\frac{2I_{xy}}{I_{xx} - I_{yy}} = -\frac{2(-1.05 \times 10^6)}{(2.45 - 1.35) \times 10^6} = \frac{2.10}{1.10} \approx 1.909$$

$$2\theta_p = \tan^{-1}(1.909) \approx 62.35^\circ \implies \theta_{p1} \approx 31.18^\circ, \quad \theta_{p2} = 31.18^\circ + 90^\circ = 121.18^\circ$$

2. Principal Moments of Inertia (I_1, I_2):

$$I_{1,2} = \frac{I_{xx} + I_{yy}}{2} \pm \sqrt{\left(\frac{I_{xx} - I_{yy}}{2}\right)^2 + I_{xy}^2}$$

$$\frac{I_{xx} + I_{yy}}{2} = \frac{2.45 + 1.35}{2} \times 10^6 = 1.90 \times 10^6 \text{ mm}^4$$

$$R = \sqrt{\left(\frac{1.10}{2}\right)^2 + (-1.05)^2 \times 10^6} = \sqrt{0.3025 + 1.1025} \times 10^6 = \sqrt{1.405} \times 10^6 \approx 1.185 \times 10^6 \text{ mm}^4$$

$$I_{\max} = I_1 = 1.90 + 1.185 = 3.085 \times 10^6 \text{ mm}^4$$

$$I_{\min} = I_2 = 1.90 - 1.185 = 0.715 \times 10^6 \text{ mm}^4$$

3. Invariant Cross-Check: $I_1 + I_2 = 3.085 + 0.715 = 3.80 \times 10^6 = I_{xx} + I_{yy} = 2.45 + 1.35 = 3.80 \times 10^6$
(Confirmed!).

5 Part 5: HBTU Semester Exam Quick Reference & FAQs

High-Yield HBTU Examination Key Rules

- **Semicircle Centroid Offset:** Remember the distance from the flat diametral base to the centroid is $\frac{4R}{3\pi}$, whereas for a semicircular arc (wire/curve) it is $\frac{2R}{\pi}$. Do NOT mix up these two factors!
- **Steiner's Theorem Axis Requirement:** You can ONLY shift between an arbitrary axis and the **centroidal axis** directly via $I = I_G + Ad^2$. Shifting between two arbitrary non-centroidal axes requires a two-step process: $I_{\text{arbitrary } 1} \rightarrow I_G \rightarrow I_{\text{arbitrary } 2}$.
- **Mass MOI Coefficients Summary:** Ring = MR^2 , Solid Disc = $\frac{1}{2}MR^2$, Solid Sphere = $\frac{2}{5}MR^2$, Spherical Shell = $\frac{2}{3}MR^2$, Cone = $\frac{3}{10}MR^2$.
- **Product of Inertia Symmetry:** Whenever a shape is symmetric about either the X or Y axis (e.g., I-section, T-section, channel about web center), $I_{xy} = 0$, meaning the geometric axes are already the principal axes!